

2 (a) All organisms respire to release energy from energy-rich molecules such as glucose.

(i) Anabolic reactions **use** energy.

State **two** examples of anabolic reactions occurring in living organisms.

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(ii) Describe **and** explain the use of ATP in glycolysis.

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(b) Some factory workers in the early 20th century were exposed to chemical X and experienced serious side-effects.

Some of the effects of exposure to chemical X are:

- decreased production of ATP
- increased lipid metabolism, with weight loss
- increased production of pyruvate and lactate
- excess heat energy release, causing an increase in body temperature, which can be fatal.

Chemical X increases the permeability of the inner mitochondrial membrane to protons (hydrogen ions), causing some protons to leak out into the matrix.

(i) Explain why people exposed to chemical X show:

- decreased production of ATP
- increased lipid metabolism.

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(ii) Suggest **and** explain why chemical X causes increased production of pyruvate and lactate.

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(iii) Suggest why excess heat energy is released.

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[Total: 11]